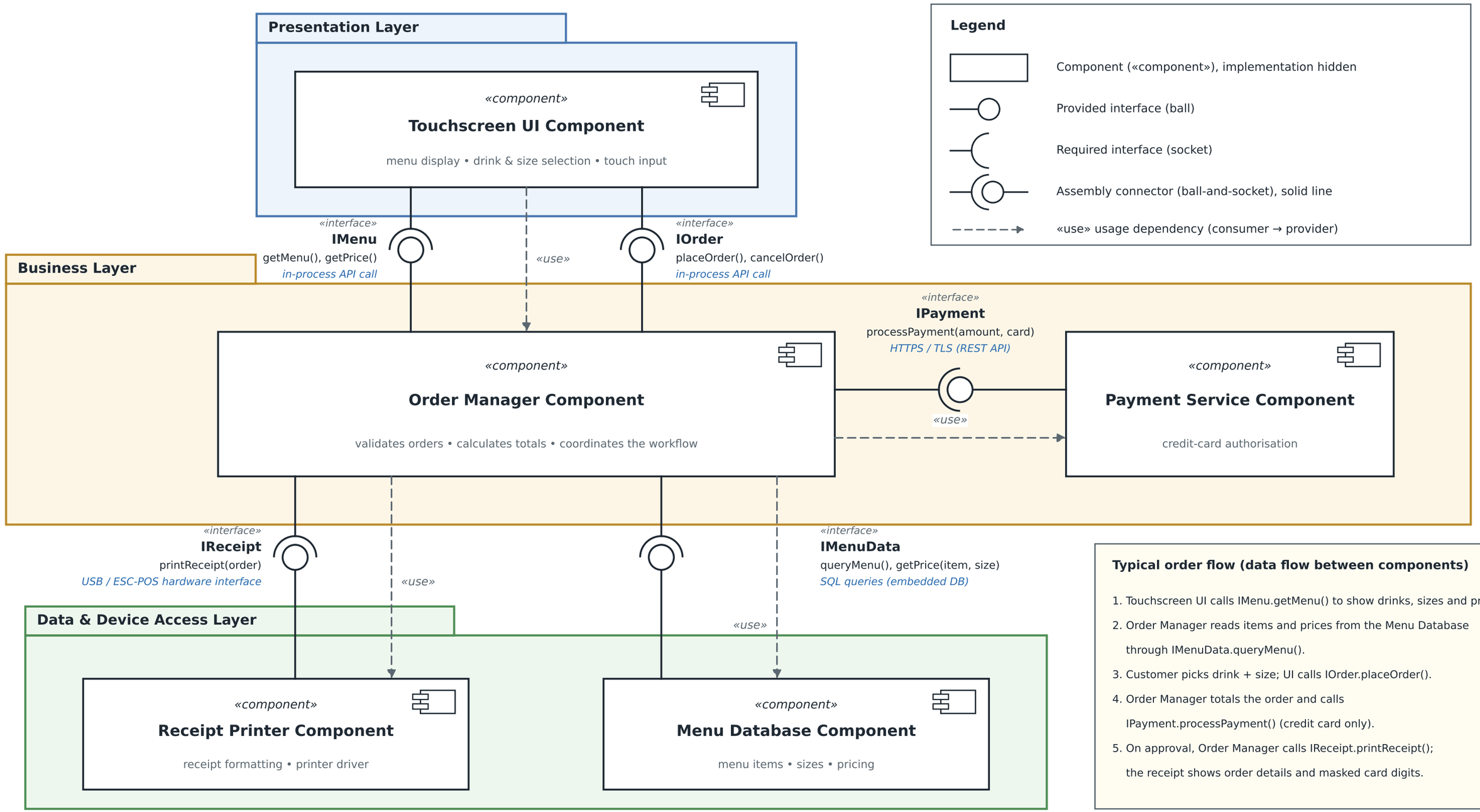
Self-Service Coffee Kiosk System – Layered Architecture

UML Component Diagram | Lab 3: Component Modelling & Architectural Pattern Selection



Legend

- Component («component»), implementation hidden
- Provided interface (ball)
- Required interface (socket)
- Assembly connector (ball-and-socket), solid line
- «use» usage dependency (consumer → provider)

Typical order flow (data flow between components)

1. Touchscreen UI calls IMenu.getMenu() to show drinks, sizes and prices.
2. Order Manager reads items and prices from the Menu Database through IMenuData.queryMenu().
3. Customer picks drink + size; UI calls IOrder.placeOrder().
4. Order Manager totals the order and calls IPayment.processPayment() (credit card only).
5. On approval, Order Manager calls IReceipt.printReceipt(); the receipt shows order details and masked card digits.